

Semantic Communication for Edge AI

Can a neural model learn what an edge AI task needs to communicate—without sending every pixel perfectly?



THESIS

Train a residual neural encoder and dual-head decoder end to end through a differentiable noisy channel; classification provides the measurable AI task.

A simulation-first capstone with an optional real-radio demonstration later.

The AI is only as useful as the link feeding it

Imagine a remote camera sending images to a server that decides what it sees.

REAL-WORLD TENSION

- 1** The sensor is constrained Bandwidth, latency and energy are limited at the edge.
- 2** The wireless link is imperfect Noise can corrupt a short message or make decoding fail.
- 3** The receiver has a narrow goal For classification, the final question is often simply: what is in the image?

DESIGN QUESTION

Why spend the entire link budget reconstructing every pixel if the receiver ultimately needs a task decision?

The project tests this question; it does not assume the learned method must win.

AI/ML goal: learn a compact, channel-robust representation that preserves downstream inference accuracy.

From bit-perfect delivery to learned AI representations

CONVENTIONAL COMMUNICATION Compress the image → protect the bits → reconstruct the file → classify it

Primary objective: recover source bits and pixels

NEURAL TASK-ORIENTED LINK Residual encoder → differentiable noise → dual-head decoder → class logits

Primary objective: learn a channel-robust representation for inference

EXPECTED SIGNATURE

As the link worsens, a separated chain may hit a decoding cliff; a learned joint system may degrade more gradually.

IMPORTANT CONTROL

A digital system can also send learned features—so the experiment includes that third arm.

Research question: does the task-oriented joint design help after bandwidth, noise and comparison strength are matched fairly?

How AI/ML and communication research lead here

Finite blocklength

[1–4]

Rate backs off from capacity; source and channel dispersions interact.

OPEN ISSUE

Need measured practical short-packet loss.

Learned compression

[11–17]

End-to-end transforms and entropy priors improve rate–distortion.

OPEN ISSUE

Reconstruction optimization is not task optimization.

DeepJSCC

[5–10, 24]

Continuous learned mappings exhibit graceful degradation under mismatch.

OPEN ISSUE

Most image studies foreground reconstruction quality.

Task-oriented communication

[18–23]

Learned representations can preserve remote inference under link constraints.

OPEN ISSUE

Task awareness must be separated from joint coding.

Gap carried forward: separate neural representation learning from joint channel coding in a controlled three-way comparison.

What we will build—and what counts as success

OBJECTIVES

1. Train a residual neural encoder and dual-head decoder through AWGN.
2. Implement a standards-derived, non-strawman JPEG 2000 + NR LDPC baseline.
3. Tune classical quality, code rate and modulation on validation only.
4. Add a shared-encoder digital AI feature control for attribution.
5. Evaluate paired image outcomes once on the sealed test split.
6. Report positive, null or negative outcomes under one frozen protocol.

H1 · PRIMARY

Low-SNR separation: learned – classical_adaptive

A point qualifies when $\sqrt{N} \cdot \Delta(s) / \sigma(s) > 1.96$. H1 is supported only if the longest run R_{obs} contains at least three consecutive qualifying points at or below the training SNR and the calibrated run p-value ≤ 0.05 .

Effect size of record: mean paired accuracy difference over the full low-SNR region.

H2 Graceful vs cliff

Fixed classical curve drops ≥ 30 pp while learned drops ≤ 15 pp over the frozen window.

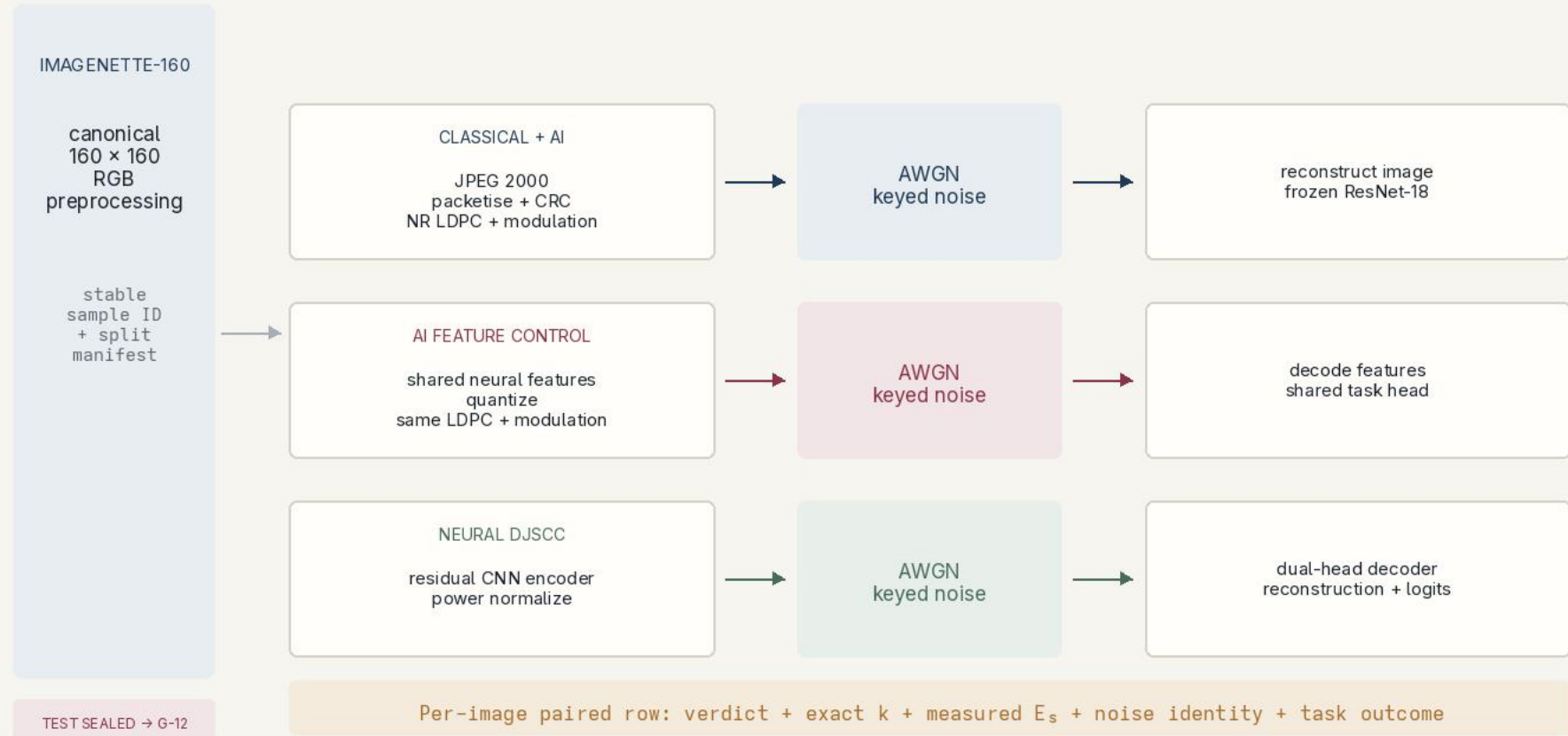
H3 Convergence

Paired gap trends toward zero across SNR; crossover is reported, not required.

H4 Attribution

Credit joint coding only if DJSCC also exceeds the task-aware digital control under the H1 rule.

AI/ML architecture and three-arm experiment



How we keep the comparison fair

Equal resource

Same complex-symbol budget k and E_s/N_0 definition; realized symbol energy is logged.

Strong comparator

JPEG 2000 quality, LDPC rate and BPSK/QPSK/16-QAM tune per SNR on validation.

Exact accounting

Container, CRC, code-block and filler bytes count; decode failures remain in the denominator.

Measured BLER

Every required physical identity is characterized; missing evidence is never treated as zero BLER.

Outage policy

Constant-class fallback is measured from validation counts—100/1,000—not assumed as 1/10.

Classifier control

Clean and artifact-finetuned classifier passes separate codec shift from link failure.

Attribution arm

Quantized learned features traverse the same digital chain, isolating task awareness from joint coding.

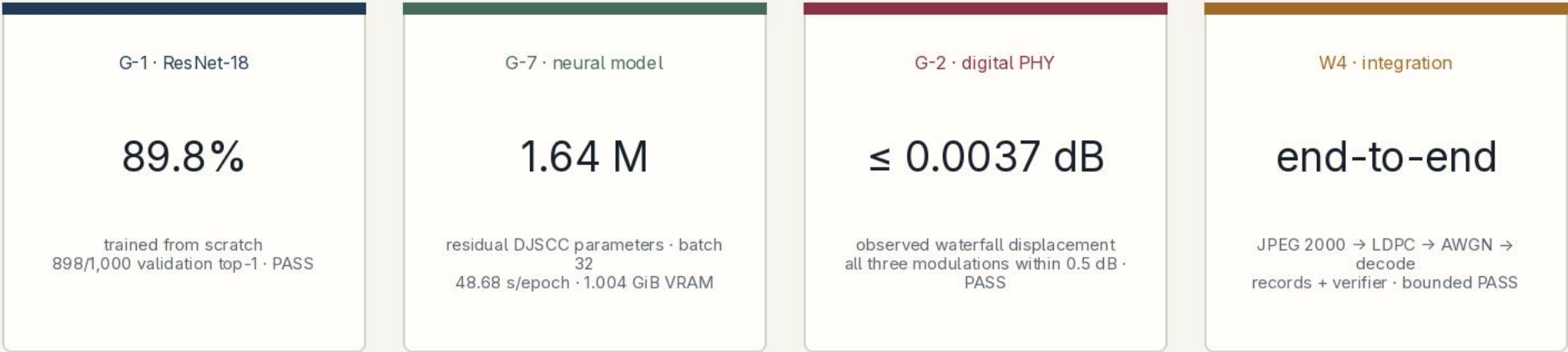
One test campaign

All choices freeze before G-12; paired inference retains complete image × system trajectories.

Fail-closed rule: uncharacterized, infeasible and failed cases are explicit outcomes—not silently dropped rows.

AI/ML feasibility and system evidence

NO HEADLINE COMPARISON EXISTS YET · bounded smoke is integration evidence, not scientific outcome evidence



G8_C · CURRENT

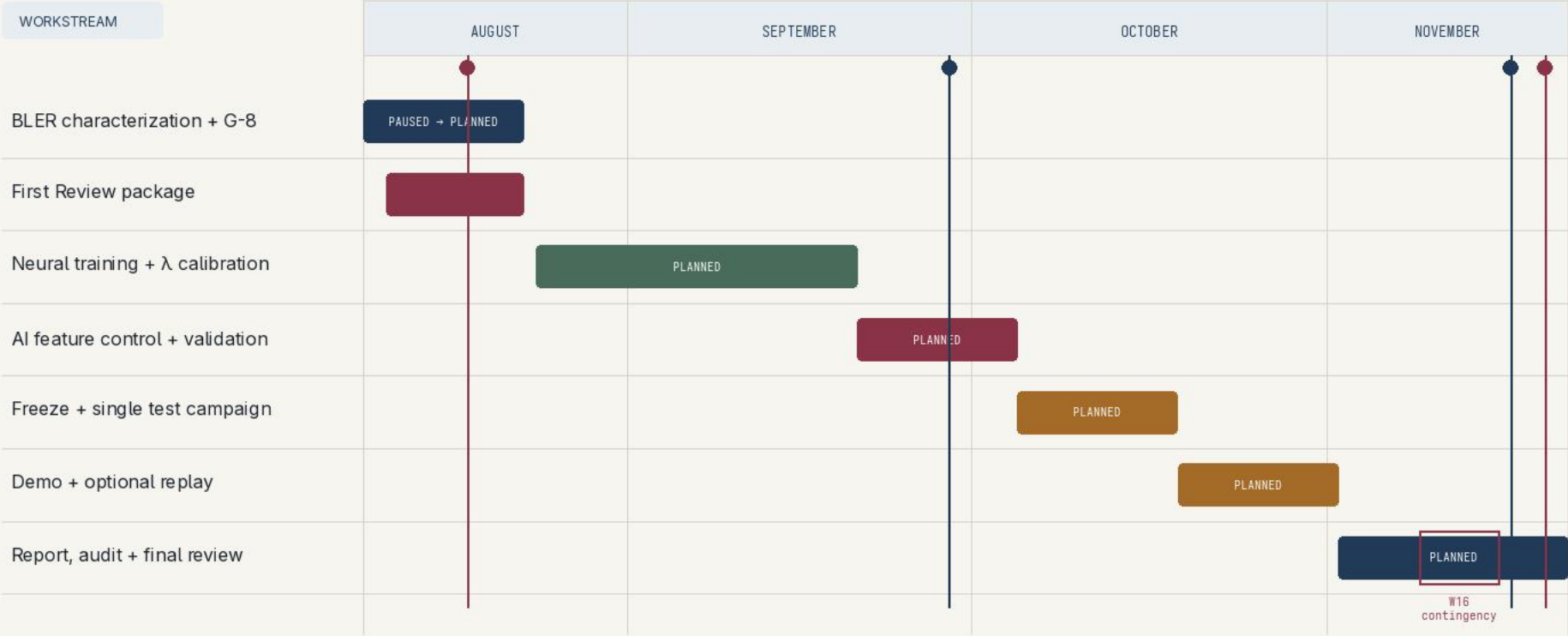
C2 paused

authenticated durable checkpoint
source epoch 2 registered · safely resumable

BLER table: not frozen
selection / inference / training / test: 0

Interpretation: the reference task, computational budget and digital chain are verified; the comparison itself remains ahead.

Project Gantt chart



Review dates: 18–22 Aug · 29 Sep–3 Oct · 17–21 Nov · report due 20 Nov

W16 is report contingency—not room for new experiment scope.

Edge AI application, AIML contribution and scope

APPLICATION

Edge AI application

Remote cameras and sensors may need receiver-side inference over limited, noisy links.

The sender runs a compact encoder; the receiver owns the decoder and task head. This represents split edge/cloud inference.

The measured task is Imagenette-160 classification, but the method is neural representation learning for communication.

IMPLEMENTATION

Deployment ladder

TIER 1 · required
Simulation-first, offline and reproducible.

TIER 2 · stretch after G-5
Conducted SDR I/Q replay; prerecorded outcome expected.

TIER 3 · stretch
Edge placement only if latency and procurement gates pass.

Candidate: HackRF One + attenuation chain + RTL-SDR; no purchase yet.

METHOD

AI/ML contribution

Residual CNN encoder with GroupNorm and PReLU.

Dual-head decoder for class logits and reconstruction.

Supervised multi-task loss: cross-entropy + λ MSE.

Differentiable AWGN enables end-to-end training.

The digital feature arm isolates representation learning from joint coding.

TECHNICAL BOUNDARY

OpenJPEG 2.5.4 + TS 38.212-derived LDPC/rate matching over abstract AWGN—not a full 5G NR link claim

No guaranteed win, measured energy saving or real-radio generalization; completion means executing the preregistered protocol.

Explicit First Review rubric-to-slide map

CRITERION	SLIDES	EVIDENCE EXPOSED IN THE DECK
Motivation	2–3	Short packets, fixed k , task accuracy, finite-blocklength costs
Objectives	3, 5–7, 10	Neural method, three-arm build, fair protocol and attribution control
Hypothesis	5	Exact H1 run rule; H2–H4 preregistered; crossing not required
Problem Survey	2, 4, 12	30-source synthesis across four literature families
Subject Knowledge	3–8, 10	Residual CNN, multi-task loss, AWGN, controls, evidence and scope
Time Plan	9	Gate-ordered critical path and fixed review/report dates

Final PASS still requires the real PPT review, four-member rehearsal, guide acknowledgement and annotated review-1-basis tag.

Selected references and evidence provenance

SELECTED LITERATURE

[1] Shannon, A Mathematical Theory of Communication, 1948.

[2] Gastpar, Rimoldi & Vetterli, To Code, or Not to Code, 2003.

[3] Polyanskiy, Poor & Verdú, Finite Blocklength, 2010.

[4] Kostina & Verdú, Lossy JSCC at Finite Blocklength, 2013.

[5] Boursoulatz, Kurka & Gündüz, DeepJSCC, 2019.

[6] Kurka & Gündüz, DeepJSCC-f, 2020.

[7] Kurka & Gündüz, Bandwidth-Agile DeepJSCC, 2021.

[11] Ballé, Laparra & Simoncelli, Learned Compression, 2017.

[12] Ballé et al., Scale Hyperprior, 2018.

[17] Blau & Michaeli, Perception-Distortion, 2018.

[18] Jankowski, Gündüz & Mikołajczyk, Wireless Retrieval, 2021.

[19] Shao, Mao & Zhang, Task-Oriented Edge Inference, 2021.

[20] Xie et al., Deep Learning Enabled Semantic Communication, 2021.

REPOSITORY EVIDENCE

G-1	results/reference_classifier/g1_adjudication.json 89.8% validation; test sealed
G-7	results/profiling/g7_djscc_profile.json 1.64M params; measured epoch/VRAM
G-2	results/baseline/g2/g2_adjudication.json golden vectors + BLER conformance
W4	results/baseline/w4/integration_adjudication.json bounded end-to-end integration
G8_C	results/baseline/g8/campaign_state.json authenticated characterization cursor
Spec	spec/SPEC.md normative hypotheses and requirements
Plan	docs/gantt-plan.md dates, gates and current status
Deploy	docs/deployment-dossier.md simulation-first + SDR stretch

Refresh G8_C status from instructions/RESUME.md on submission day.